

Explainable Forecast of Electricity Market Price Using Machine Learning Algorithms:

A Case Study of the Turkish Day-Ahead Market

Abstract

Electricity day-ahead markets facilitate trading electricity, determining a market-clearing price at the intersection of supply and demand for a specific hour. Accurate electricity price forecasting (EPF) is crucial for electricity markets, enabling more competitive bids from wholesale sellers and buyers, which can lower prices for end-users and enhance social welfare. Model explainability is also critical in EPF. It promotes transparency and trust in the models and empowers stakeholders to make informed decisions and manage risks effectively. This study evaluates the state-of-the-art machine learning methods for forecasting the Turkish day-ahead electricity prices. The XGBoost, CatBoost, LightGBM, Random Forest, Decision Tree, Support Vector Machine, and Sequential Neural Network models are employed with 32 independent features. Model codes and data are shared open access. The LightGBM model emerges as the most effective algorithm. We conducted statistical and SHAP value analysis for model explainability and feature importance analysis. Results indicate that historical market prices significantly influence predictions across all models. Renewable energy sources tend to lower the electricity price, while fossil fuels increase it. Economic factors like exchange rates and natural gas prices also strongly affect market dynamics. The comprehensive analysis reveals the dynamic interactions between features and their impact on pricing strategies.

Keywords: Electricity Price Forecasting, Explainable Machine Learning, Market Clearing Price, Day-Ahead Market, SHAP.

1. Introduction

Electricity markets are central to modern energy systems, fostering competition, efficiency, and investment. Well-functioning markets drive economic growth and attract investment crucial for reliable energy infrastructure. The market clearing price (MCP), determined via daily auctions, is pivotal in aligning supply and demand. Through bid submission, generators and retailers offer electricity volumes at specific prices. Lower generation bids are prioritized for dispatch, ensuring grid-connected power generation, while higher bids may be rejected (Figure 1). The market clearing price (MCP) significantly impacts both generators and consumers. It determines generator revenue and consumer costs. MCP serves as a crucial signal, guiding pricing, influencing market behavior, and shaping investment decisions. Accurate MCP forecasts, derived from historical data and appropriate models, are essential for effective bidding and pricing strategies. Electricity price forecasting is paramount for informed decisions, risk management, and efficient market operation, including demand response, policy assessment, and renewable energy integration.

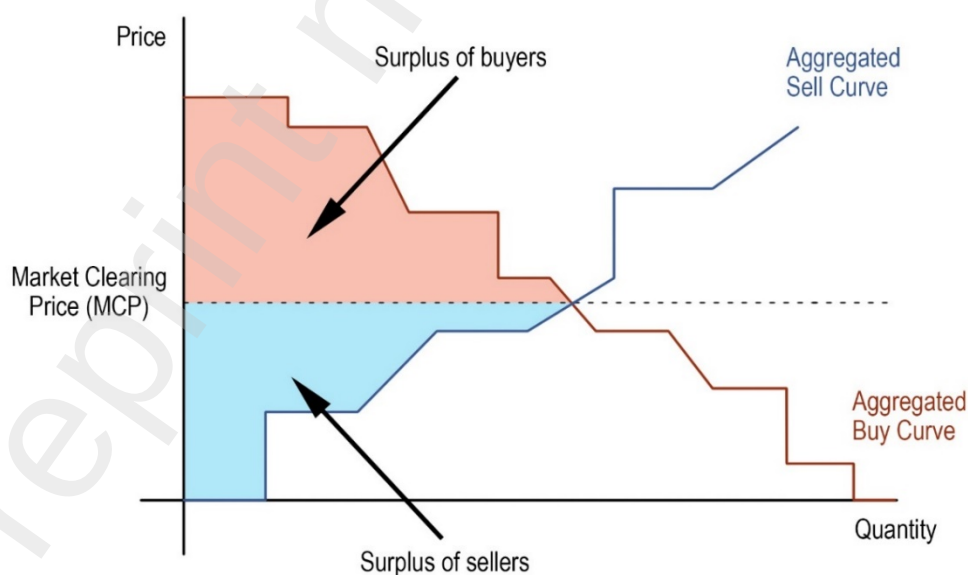


Figure 1. Market Clearing and MCP

A considerable number of studies exist concerning electricity price forecast (EPF), encompassing a variety of models [1-4]. Researchers such as Niimura [5], Tafreshi [6], Ludwig et al. [7], and Weron [8] have reviewed various EPF techniques and models developed over different years. The reviewed studies categorize these techniques and models into time series, simulation models, game theory-based models, and statistical models, further dividing them into stationary and non-stationary time series models. EPF models can be differentiated into linear and non-linear models, with linear models typically exemplified by statistical models, while non-linear models are represented by approaches such as machine learning and deep learning. Linear models offer a practical structure for model prediction, whereas non-linear models tend to yield more precise outcomes [9-11].

Statistical time series models often include methods that perform adequately during active market periods, aiming to model price fluctuations through autoregressive methods and conditional heteroscedastic models [10, 12]. Frequently utilized models for EPF include the Autoregressive (AR) model [11, 13, 14], Integrated Moving Average (ARIMA) [13, 15-18], Autoregressive Moving Average (ARMA) [13, 19, 20], and Seasonal Integrated Moving Average (SARIMA) [21].

Several studies have examined the performance of models within the scope of EPF using the LSTM technique, yielding approximately similar results [3, 8, 22-30]. Some research has employed deep neural network models, including Feedforward Neural Networks (FNN) [31, 32], Recurrent Neural Networks (RNN) [3], Convolutional Neural Networks (CNN) [3, 30, 33], and Artificial Neural Networks (ANN) [8, 13, 34-41], taking into account parameters (price formation, volatility, exogenous variables, etc.) affecting MCP to determine an appropriate training matrix and conduct analyses for optimal price

prediction. For instance, Mehrdoust et al. [40] employed an ANN-based model for Nordic EPF, optimizing parameters via genetic algorithms.

For machine learning (ML) models in EPF, Support Vector Machines (SVM) [8, 13, 16, 42-44], Random Forest (RF) [28, 29, 39, 43, 45], Multiple Linear Regression (MLR) [30, 46], XGBoost Regression [30, 32, 39, 47] and Decision Trees Regression (DTR) have been utilized. Comparative articles on statistical methods and machine learning approaches exist, with Uniejewski et al. [48] arguing that statistical methods outperform ML models. However, across the literature, ML models are predominantly defended for exhibiting higher performance in prediction, although results of similar studies vary based on data set characteristics [22, 49]. According to Uniejewski et al. [50], Lasso and elastic nets show the best performance in EPF, with elastic nets outperforming other automatic selection methods. Recent years have seen a growing importance of probabilistic EPF due to the increased uncertainty arising from smart grids and the integration of renewable energy, as highlighted by Nowotarski et al. [51].

Acaroğlu and Marquez [52] have discussed research conducted over the past two decades on electricity price and load forecasting, particularly focusing on approaches derived from wind energy, and discussing forecasting methods and variables of interest to market participants and decision-makers. The diversity of variables and methodologies for accuracy in price and load forecasts has been examined. Liu et al. [53] address high-dimensional EPF with SHAP for feature selection and LSTNet quantile regression for probabilistic forecasts, capturing uncertainty and redundancy. Studies by Gholipour et al. [54], Atef and Eltawil [55], and Kuo and Huang [56] have compared the effectiveness of deep learning in forecasting to machine learning, frequently used as a predictive tool in recent periods. While there is no consensus in the literature on which method is best for

EPF, reviews conducted to fill this gap have deduced that deep learning slightly outperforms machine learning in terms of effectiveness [55, 57].

Energy Exchange Istanbul (EXIST), established in 2015, in Türkiye, operates with a commitment to the efficient, transparent, reliable, and sustainable development of energy markets. EXIST oversees the Day-Ahead Market (DAM), Intra-Day Market, and Balancing Power Market, along with the Spot Natural Gas Market, Futures Electricity Market, Futures Natural Gas Market, and the Renewable Energy Resource Guarantee System and Market. Transactions in the DAM are conducted on an hourly basis on a daily scale. The DAM process is shown in Figure A. 1 in the appendix. Several studies have been conducted on the prediction of electricity prices in Türkiye. Yıldırım et al. [58], Özyıldırım and Beyazıt [59], Hayfavi and Talaslı [60] are among the pioneering works in this field. Kölmek and Navruz [61] presented a study for spot electricity market price prediction in Türkiye, using the first neural network approach. They employed Artificial Neural Networks (ANN) and AutoRegressive (AR) models for day-ahead price forecasting. Yıldırım et al. [58] utilized Conic Multivariate Adaptive Regression Splines (CMARS), dynamic regression, and Robust CMARS methods. The accuracy of results is uncertain due to data limitations imposed by the methodology. Hayfavi and Talaslı [60], on the other hand, proposed a multifactorial model using the average spot price, offering it as an alternative to traditional methods. However, the parameters and assumptions in the model do not fully capture the dynamics of the Turkish market. Therefore, the use of more specific and suitable models for the Turkish day-ahead electricity market is needed. Uğurlu et al. [62] emphasized the significance of temperature, utilizing linear regression and radial basis function. They found that variables such as the square of temperature, temperature, holidays, hourly, daily, and monthly prices were useful, with the radial basis function yielding better results. Avcı et al. [63] employed Seasonal Autoregressive

Integrated Moving Average (SARIMA) and ANN methods, using various variations of MCP as variables. Taysi et al. [64] aimed to evaluate price prediction systems for the Turkish electricity market, creating two different systems, an ARIMA model and a feedforward artificial neural network (ANN). Both systems predict electricity prices weekly with an average error rate of 8.5%. Ozozen et al. [65] proposed a novel ARIMA-ANN hybrid model. This approach addresses partially liberalized markets with incomplete data and high historical dependence. They report a 4% reduction in prediction errors compared to their benchmark. Table A. 1 (in the appendix) summarizes the studies on the EPF specific to the Turkish case, with detailed information and supplementary data. A review of Turkish electricity market research reveals a scarcity of recent, published articles. Most studies are doctoral theses or conference proceedings, often predating the EXIST market and lacking comprehensive data. Critically, there is a gap in research that reflects current market conditions and leverages recent data for accurate EPF. Key shortcomings include limited analysis of renewable energy integration, lack of open-source data and methods, and more importantly, clear identification and quantification of price-influencing variables and model explainability. In essence, there is a significant deficit in research that captures current market dynamics and offers effective EPF, especially for the Turkish market.

In this study, a comparison of the state-of-the-art models is presented for EPF using DAM data available on the EXIST platform. The dataset covers a period of seven years from January 2015 to December 2022. This dataset captures various factors affecting Turkish electricity prices (MCP), bidding activity (volumes), generation details, weather, and fuel prices. In total, 33 variables are utilized for analysis. Random Forest, Decision Tree, Support Vector Machine, CatBoost, XGBoost, LightGBM, and Sequential Artificial Neural Networks are employed using Python programming. We utilize diverse metrics (MAE, MSE, RMSE, MAPE, R-

squared) to assess errors across various methods. The analysis leverages a rich dataset with hierarchical features, aiming to minimize prediction errors. Additionally, correlation analysis, feature engineering, hyperparameter optimization, and SHAP-based explainability are conducted. Notably, the data and code are open source, promoting transparency and reproducibility¹.

This article is structured as follows: the next section describes the methodology, summarizing the utilized machine learning models and error metrics. The third section demonstrates the dataset and model application. The fourth section presents the EPF results. The fifth section provides model explainability, a detailed explanation of features that affect the electricity price. The last section discusses the highlights of the study and concludes the article.

2. Methodology

Machine learning (ML), a subfield of artificial intelligence, equips computers to learn from data and make predictions. By training algorithms to identify patterns and improve performance, ML fosters autonomous learning and decision-making. Applications range from image recognition to natural language processing. Encompassing various subfields (**Error! Reference source not found.**), ML leverages diverse techniques for specific data analysis tasks. In this study, models such as Random Forest, Decision Tree, Support Vector Machine, CatBoost, XGBoost, LightGBM, and Sequential Artificial Neural Networks are utilized. Hyperparameter optimization for the models is achieved using the Keras Tuner.

¹ Github link for the codes and the dataset: <https://github.com/TheEmgame/EPF-Turkish-Day-Ahead-Market.git>

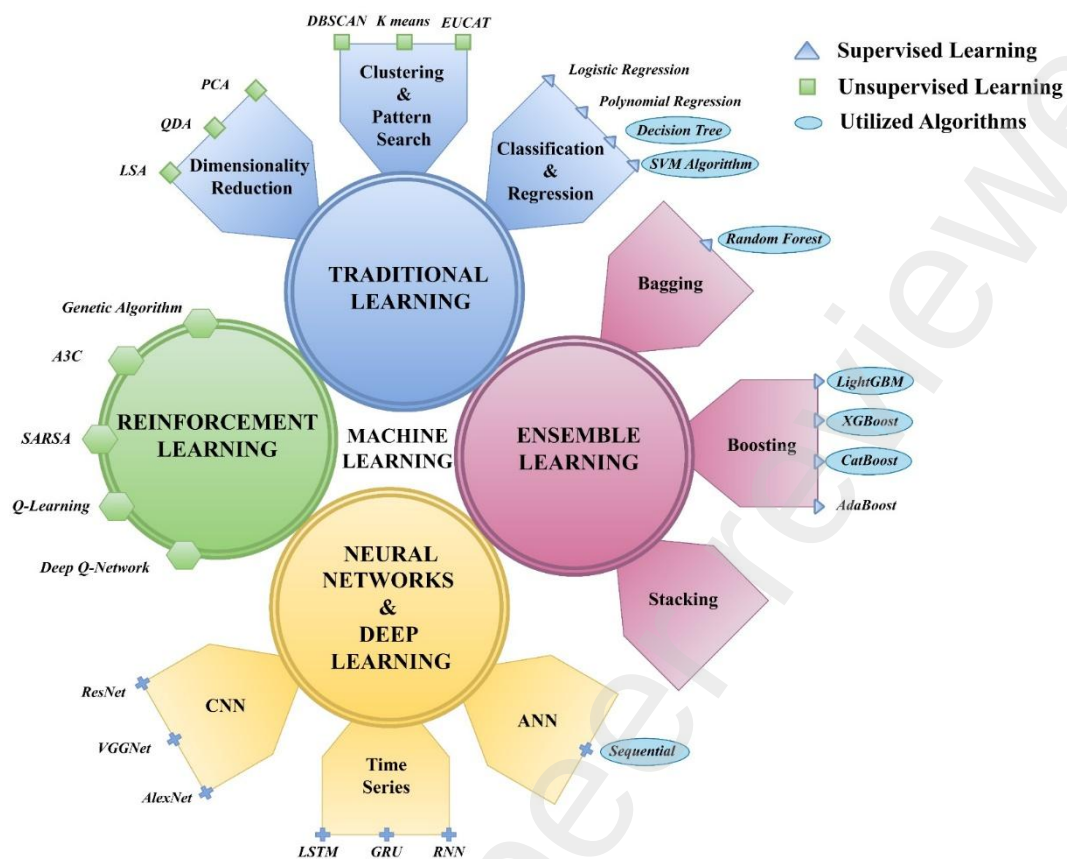


Figure 2. Machine learning models and utilized algorithms in this study

Figure 3 summarizes the methodology used for EPF by processing the data collected from EXIST through exploratory data analysis and visualizing the findings obtained. The analysis reveals the structural characteristics and basic statistics of the datasets, and effective prediction models are developed using various machine learning algorithms.

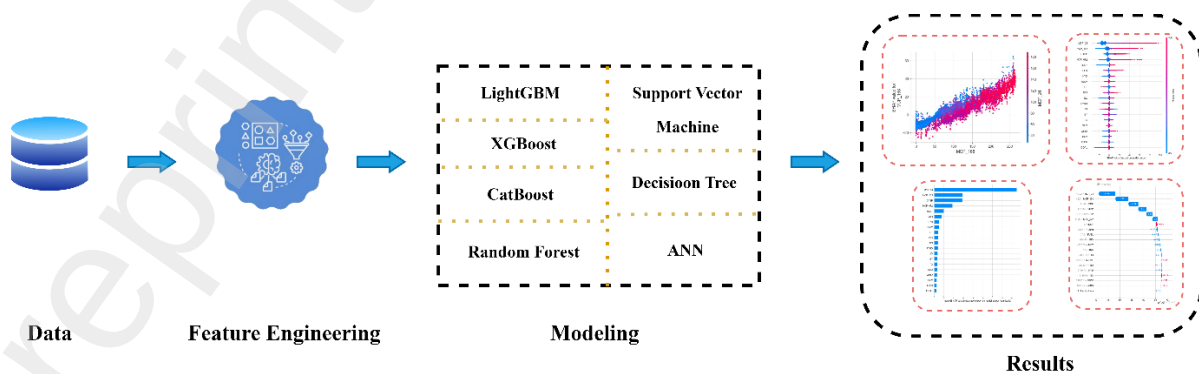


Figure 3. Methodology

Random Forest Regression, a supervised learning method, offers advantages in speed, ease of use, and handling of multidimensional data compared to boosting techniques, while mitigating overfitting issues prevalent in decision trees [66]. Support Vector Machines (SVM), a supervised learning method, utilizes kernel functions for non-linear regression in high-dimensional space [67, 68]. Extreme Gradient Boosting (XGBoost), an open-source library for gradient boosting, leverages decision trees for regression tasks. Unlike random forests, XGBoost employs a weighted linear sum of trees to minimize residual errors, aiming for improved predictions [69, 70]. LightGBM, an open-source gradient boosting library, prioritizes large gradients and utilizes automatic feature selection for faster training and enhanced prediction [71]. CatBoost Regressor uses a target-based algorithm to reduce overfitting and train on the entire dataset. It addresses noise in low-frequency categories and traditional boosting issues by introducing random permutations to evaluate leaf values, preventing overfitting [72]. Decision tree regression is a simple, interpretable algorithm that captures non-linear relationships. It segments data hierarchically, with branches representing features and thresholds. Data points follow paths from the root to terminal nodes, where the mean value predicts the target. Despite its advantages, it can overfit and be slow on large datasets [73].

Keras is a high-level API for deep learning, utilizing TensorFlow for model creation and training. It offers a wrapper class for building sequential neural networks and provides methods for compilation, training, and evaluation. The Keras model structure leverages TensorFlow during the compilation phase². Keras is used in this study because it enables the rapid and straightforward development of deep learning models. The Sequential model in Keras allows for modeling past data to predict future prices. Utilizing TensorFlow, Keras offers high-

² www.tensorflow.org

performance model compilation, training, and evaluation. Methods like hyperparameter optimization and cross-validation are easily implemented, enhancing model accuracy. Therefore, Keras is an effective and efficient tool for EPF projects.

Five error metrics are used to evaluate model performance. These are R^2 , MSE, MAE, RMSE, and MAPE. R-squared (R^2) is the square of the correlation coefficient, measuring the proportion of the dependent variable's variation explained by the independent variables in the model. It indicates the strength of the relationship between independent and dependent variables on a scale from 0 to 1, where a value of 1 represents perfect correlation. R^2 does not consider the issue of overfitting. It is calculated as follows, where $SS_{Residual}$ is the sum of squared errors, and SS_{Total} is the sum of mean differences:

$$R^2 = 1 - \frac{SS_{Residual}}{SS_{Total}} \quad \#(1)$$

Mean Squared Error (MSE) measures how close a regression curve is to a set of points. It is always a positive value and a lower MSE value indicates better performance of the model. MSE is calculated as follows where y_i represents the actual values, $\hat{y}_i$ represents the predicted values, and N is the number of samples:

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad \#(2)$$

Mean Absolute Error (MAE) is a measure of the difference between two continuous variables. MAE, for each example in the dataset, is the average of the absolute differences between the actual value and the predicted value. It is commonly used in regression and time series problems and is calculated as follows:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad \#(3)$$

Root Mean Square Error (RMSE) is a commonly used metric in machine learning to measure the distance between the values predicted by a model and the actual values. It is a second-degree metric that quantifies the magnitude of the error. RMSE represents the standard deviation of prediction errors (residuals). Residuals are a measure of how far the regression line is from the data points. RMSE indicates how spread out these residuals are around the line that best fits the data. It informs about the density of the data points around the line that fits them the best. The RMSE value can range from zero to infinity, with lower values indicating better performance of the predictor. An RMSE value of zero implies that the model makes no errors. RMSE is the square root of Mean Squared Error (MSE), and it is calculated as follows:

$$RMSE = \sqrt{\frac{\sum_{j=1}^n (y_j - \hat{y}_j)^2}{n}} \#(4)$$

Mean Absolute Percentage Error (MAPE) is an error metric that utilizes absolute values to prevent positive and negative errors from canceling each other out. It is commonly used in time series models to compare prediction accuracy. MAPE values below 10% indicate high accuracy in prediction models, while models with MAPE values between 10% and 20% can still be classified as accurate. MAPE is calculated as follows:

$$MAPE = \frac{\sum_{j=1}^N \frac{|y_j|}{\hat{y}_j}}{N} \#(5)$$

These are common evaluation metrics used in ML to assess the performance of models, particularly in the context of regression problems. Each metric serves a different purpose, and the choice of which one to use depends on the specific characteristics of the problem and the emphasis of the evaluation. R^2 is used to understand the goodness of the fit. MSE and RMSE are used to measure error magnitude, with RMSE providing an error measure in the same units as the target variable. MAE is used when a simple and interpretable error measure that is less

sensitive to outliers is desired. MAPE is used when an assessment prediction accuracy in percentage terms is needed, often in forecasting scenarios. We applied all the above measures so that comparisons with other studies could be available when needed.

3. Dataset and Model Application

In this study, a comparison of forecasting methods for the Turkish Day-Ahead Market (DAM) clearing price is presented. The algorithms generate a forecast for the hourly Market Clearing Price (MCP) for 24 hours of each day using the data presented in Table 1. The data consists of several features collected from the Energy Exchange Istanbul (EXIST) Transparency Platform³, Petroleum Pipeline Corporation, and the General Directorate of Meteorology. The dataset covers the duration from January 1, 2015, to December 31, 2022. The MCP feature in the table is the observed MCP of the hour which is the dependent variable of the algorithms. The rest are independent variables that are used for prediction.

³ seffaflik.epias.com.tr

Table 1. Data Set

Feature	Explanation	Mean	Std	Min	Max
MCP	Observed Market Clearing Price (MCP) of the hour (USD/MWh)	67.449	49.874	0.000	66.630
HOLY	Binary value that represents whether an hour corresponds to a weekday, weekend, or holiday	0.690	0.462	0.000	1.000
MCP-24	Observed MCP on the previous day of the prediction (24 hours before) (USD/MWh)	67.380	49.814	0.000	66.305
MCP-168	Observed MCP one week (168 hours) before the prediction (USD/MWh)	66.797	49.076	0.000	65.230
MCP-672	Observed MCP one month (672 hours) before the prediction (USD/MWh)	65.014	46.667	0.000	62.085
PISO	Price Independent Sales Order (Sum of sale order volumes at 0 price step) (MWh)	11334.934	2856.462	3721.1	23000.6
PIBO	Price Independent Bid Order (Sum of bid order volumes at price cap, 4800 TL/MWh)	18811.540	4594.161	3721.3	43135.4
SSOV	Submitted Sales Order Volume (Sum of sale order volumes at price cap, 4800 TL/MWh)	29642.208	4767.823	10444.2	44737.7
SBOV	Submitted Bid Order Volume (Sum of bid order volumes at 0 price step) (MWh)	24578.563	4623.185	10715.2	47542.8
MO	Matched Offers Quantity (MWh)	19744.961	3776.736	9640.1	32222.5
MB	Matched Bids Quantity (MWh)	19745.167	3776.621	9640.2	32223.7
TV	Trade Value (USD)	1364707.72	1074959.50	0.000	6442106.8
DER	Daily Exchange Rate (TL/USD)	8.593	4.398	3.732	18.703
TG	Total Generation (MWh)	34207.754	5208.552	15333.3	51285.6
NG	Generation from Natural Gas (MWh)	8954.022	3999.599	496.66	19453.6
HYD	Generation from Dam Hydro (MWh)	5727.298	2921.057	296.690	15891.78
LIG	Generation from Lignite Coal (MWh)	4953.004	614.350	2189.10	6984.15
RHYD	Generation from River Hydro (MWh)	2217.594	1324.942	379.960	6103.59
ICOAL	Generation from Import Coal (MWh)	6913.349	1771.430	956.350	10138.84
WIND	Generation from Wind (MWh)	2992.833	1824.538	42.530	9327.74
SOL	Generation from Solar (MWh)	114.069	245.311	-1.700	1289.82
FUEL	Generation from Fuel Oil (MWh)	84.487	59.370	0.000	237.60
GEO	Generation from Geothermal (MWh)	1021.936	177.328	525.110	1357.30
ASPH	Generation from Asphaltite Coal (MWh)	246.673	84.611	0.000	368.720
BCOAL	Generation from Black Coal (MWh)	403.430	116.198	77.500	692.620
BIO	Generation from Biomass (MWh)	523.189	219.352	202.580	1011.230
IMEX	Import-Export Electricity Volume (MWh)	-27.012	315.928	-3130.50	1648.540
WAS	Generation from Waste Heat (MWh)	82.855	13.630	30.690	127.760
GASP	Natural Gas Price (USD/1000Sm ³)	377.061	303.509	162.850	1444.510
ANK	Hourly average temperature of Ankara (°C)	13.700	9.299	-13.000	38.600
IST	Hourly average temperature of Istanbul (°C)	16.390	7.577	-2.600	35.900

3.1 Preliminary Statistical Analysis

Electricity prices in the Turkish DAM vary hourly, daily, and seasonally. Prices increase during high-demand hours and decrease on weekends and holidays when demand is lower (Figure 4). Seasonally prices surge in the winter months due to heating needs, and in the summer months due to cooling requirements. However, prices decrease in spring, possibly due to the higher share of hydroelectric power plants. General statistics and further analysis of the Turkish DAM are provided in Appendix B.



Figure 4. Average electricity prices in the Turkish Day-Ahead Market by Weekend - Holidays and Public Holidays in 2022

Correlation indicates the direction and effect of a linear relationship between variables in the dataset. Thus, by making logical inferences from the dataset, variables that do not affect predictions can be identified. The linear relationships between variables are provided as a Correlation Heat Map in Figure C. 1 in Appendix C, and the positive and negative effects of parameters influencing the closing market price are illustrated in Figure 5.

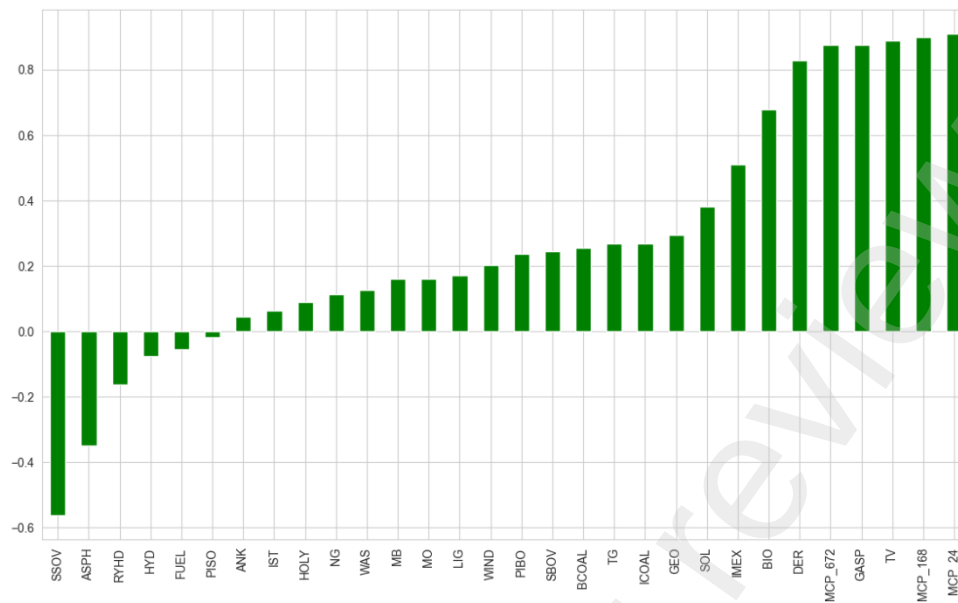


Figure 5. Independent variables affecting MCP

3.2 Data Preprocessing

To develop the most accurate model for predicting the Turkish DAM price, the following procedures are implemented:

- Normalization of the target variable and the remaining predictor variables for the required models (SVM and Sequential model).
- Preprocessing of the data, including handling missing data, removing duplicates and outliers, transforming, cleaning, normalizing, and dimensionality reduction, leading to the separation of training and test sets.
- Hyperparameter optimization for all models.
- Examination of errors in the results and selection of significant independent variables influencing the best model to analyze and reduce errors.

Data preprocessing tackles errors and missing values, and reduces data size, while normalization and scaling improve machine learning performance. Standardization (mean 0, standard deviation 1) benefits distance-based algorithms, while normalization (0-1 range) is suited for weight-sensitive models. Both input features and target variables are subjected to separate scaling procedures with hyperparameter optimization to ensure optimal model

performance. Crucially, distinct scalers and transformers are employed for input data and predicted values, further enhancing generalization.

Feature scaling is critical for machine learning models, requiring preprocessing of both input and target variables. Here, separate scalers and transformers with hyperparameter tuning are employed to optimize performance.

The data preprocessing approach involves three stages. First, data cleaning addresses structural inconsistencies and removes inappropriate data to ensure dataset integrity. Second, missing values in specific variables (MO, IST, ANK) are imputed using medians to maintain data structure. Finally, outliers are removed based on statistical analysis for a more representative dataset.

Initial trials using data between 2015-2022 reveal significant data deficiencies in PISO, PIBO, SSOV, SBOV, MO, MB, SOL, ASPH, and IMEX columns. This negatively impacts model accuracy. However, narrowing the data range to 2018-2022 significantly reduces missing data and improves prediction accuracy. This highlights the importance of appropriate timeframes in minimizing missing data and enhancing model performance, leading to more reliable and valid results.

3.3 Model Configurations

Following preprocessing, machine learning and neural network models are built. An 80-20% training-testing split is used. Min-Max scaling (normalization between 0-1) is applied to improve model convergence, particularly for SVM and ANN models, which are sensitive to feature magnitudes. Conversely, tree-based models (Random Forest, XGBoost, etc.) are not scaled as they rely on relative feature rankings and are unaffected by magnitude differences. This simplifies data preparation for these flexible models.

Seven machine learning models are evaluated in this study. Each model is subjected to a three-step optimization process. First, a baseline model is established. Second, k-fold cross-validation is implemented to assess model performance and mitigate overfitting/underfitting. Cross-validation involves iteratively splitting the data for training and testing, providing a more robust evaluation of generalization. Third, hyperparameter tuning using Grid Search is employed to identify the optimal model configuration. Grid Search systematically evaluates various hyperparameter combinations and selects the one yielding the best performance based on metrics like mean absolute error (MAE) and root mean squared error (RMSE). Normalization facilitates fairer comparisons between models and aids in selecting the algorithm best suited for the specific dataset characteristics. This comprehensive approach ensures a thorough evaluation of model performance and informs the model selection process.

The performance and training durations of various machine learning models are presented in Table 2. We observe the challenges encountered and the time-consuming nature of optimizing model hyperparameters. Particularly, the impractical aspects of automatic hyperparameter tuning methods such as GridSearch, which can take days in some cases, highlight the importance of the effectiveness and efficiency of manual tuning. These observations emphasize the critical importance of strategic management of time, cost, and performance balance in model optimization processes.

Table 2. Run times of GridSearch and SHAP Analysis

Models	SHAP uptime	GridSearch uptime
LightGBM	50 seconds	2 seconds
CatBoost	7 seconds	4 seconds
XGBoost	5 minutes	10 seconds
Sequential	6 minutes	20 minutes
SVR	3154 minutes	13 minutes
RF	10 seconds	10 minutes
DT	10 seconds	3 seconds

4. Results

This section presents the forecast performance of seven models. The error rates recorded during the training and testing phases of each model are detailed based on error metrics in Table 3 and Table 4, respectively. Evaluating these results is crucial for verifying the reliability of the model. Moreover, assessments made according to the training and test error metrics allow for the identification of potential issues such as overfitting or underfitting within the model.

Table 3. Training Set Error Metrics

Model	R ²	MAE	MSE	RMSE	MAPE
RF	0.990	2.889	24.953	4.995	20.654
SVR	0.956	6.088	108.723	10.427	43.632
XGBoost	0.996	2.238	10.744	3.278	13.827
Sequential	0.971	4.780	71.490	8.459	16.146
CatBoost	0.983	4.310	41.267	6.424	30.209
LightGBM	0.996	2.265	10.944	3.308	14.629
DT	0.950	6.367	125.300	11.194	54.997

Table 4. Test Set Error Metrics

Model	R ²	MAE	MSE	RMSE	MAPE
RF	0.934	6.612	166.431	12.901	64.849
SVR	0.942	6.921	145.590	12.066	59.077
XGBoost	0.946	6.113	134.433	11.595	52.852
Sequential	0.944	6.478	141.787	11.908	33.128
CatBoost	0.948	6.512	131.601	11.472	48.796
LightGBM	0.950	5.981	126.528	11.248	49.627
DT	0.893	8.369	269.699	16.243	97.705

The error rates of all prediction models based on the R², MSE, MAE, RMSE, and MAPE metrics are provided in Figure 6. Between the years 2018-2022, the DT model made less successful predictions compared to other models based on error metrics. Additionally, although

the CatBoost and XGBoost models demonstrate better results, the error rates and graphical analyses in Figure 6 reveal that the most successful predictions are made by the LightGBM model.

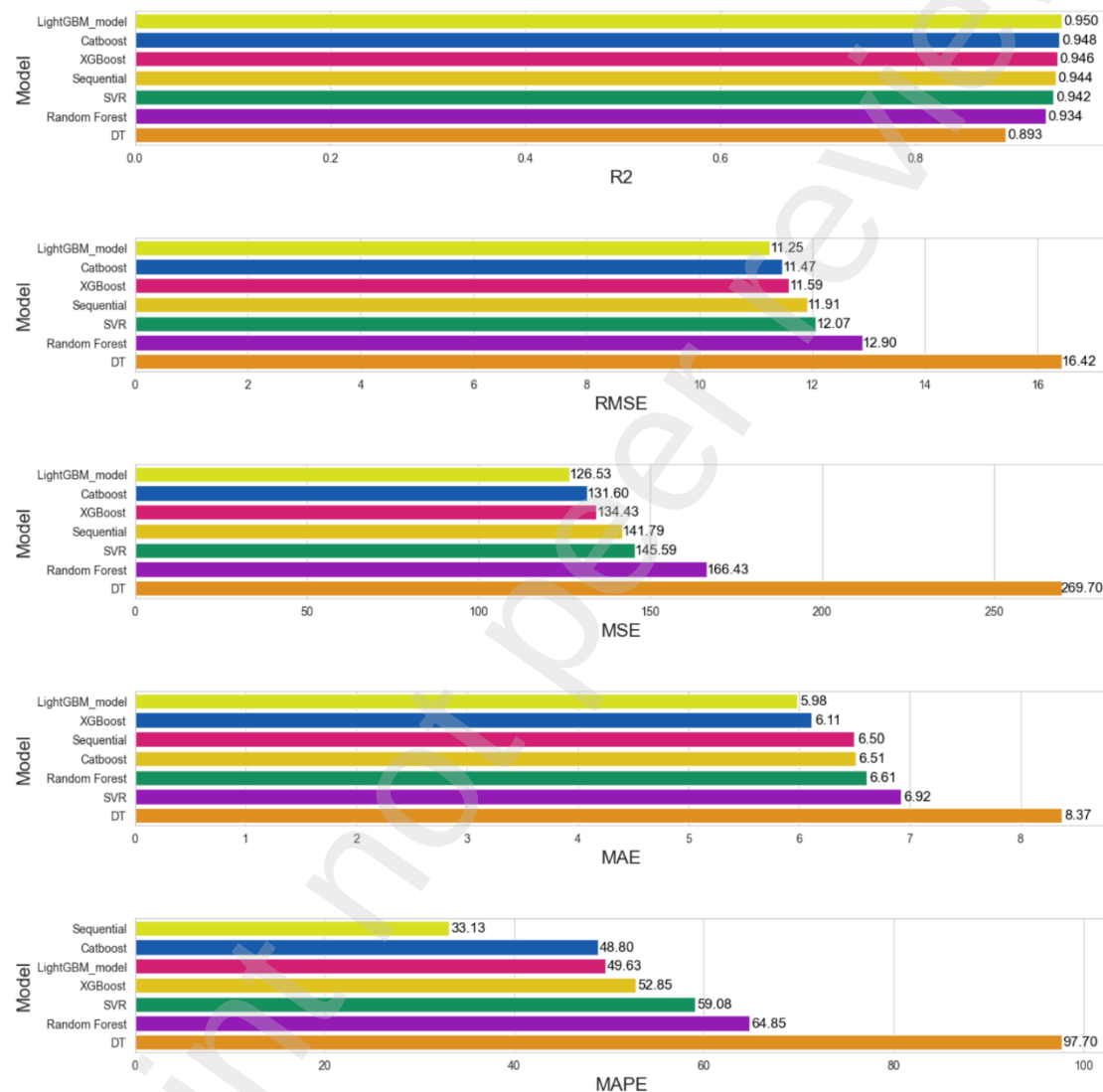


Figure 6. Result of the Models

The results are supported by the graphical analysis in Appendix D (Figure D. 1 to Figure D. 6). The LightGBM model was identified as the top-performing model, displaying an excellent alignment between actual values and predictions (Figure 7), surpassing all other models. The overlap of the black and red lines in the figure for the LightGBM model demonstrates that the model's predictions align perfectly with the actual values and indicates that this model performs

the best in EPF. The XGBoost (Figure D. 1) and CatBoost (Figure D. 6) models also stood out with their high performance, close to but slightly behind LightGBM, and can be considered strong alternatives for EPF. Further analysis is presented in Appendix D.

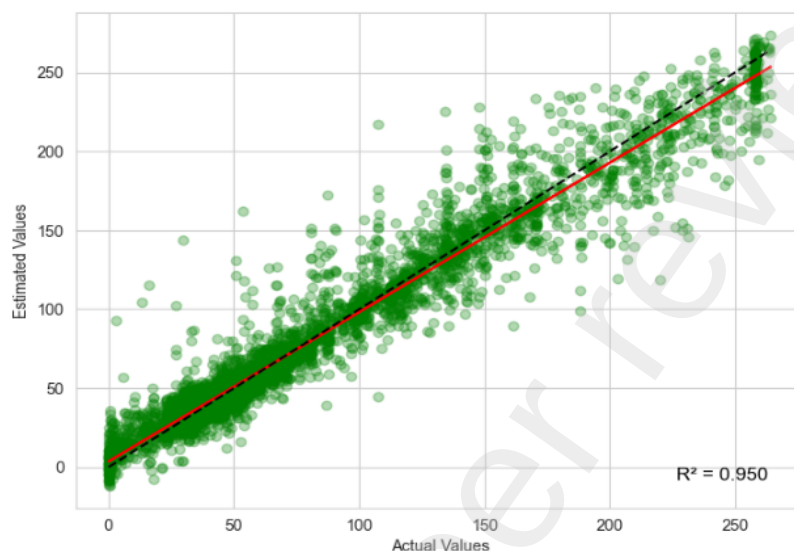


Figure 7. Prediction Error for LightGBM

5. Model Explainability

SHAP (SHapley Additive exPlanations), a concept rooted in game theory by Shapley [74], enhances the interpretability of complex machine learning models by quantifying the contribution of each feature to the model's predictions. This facilitates a deeper understanding of model behavior, feature importance, and potential biases, ultimately improving model reliability and transparency.

SHAP values provide a visual representation of the impact of individual features on model predictions. In these visualizations, color coding and the distribution of points are crucial elements for interpretation. The distribution of points along the x-axis reflects the SHAP values, quantifying the magnitude and direction of each feature's impact on the predicted Market Clearing Price (MCP). The y-axis ranks the features based on their overall importance in influencing the MCP forecast. Red points represent high feature values, while blue points represent low feature values. For instance, in Figure 8, the highest value of the MCP-24 variable

is \$264, depicted by a light red dot, and its corresponding SHAP value is +98 on the x-axis, signifying its substantial positive impact on the MCP. Conversely, the lowest value of MCP-24 is \$0, represented by a light blue dot with a SHAP value of -30, indicating a notable negative impact on the MCP.

The presence of red points with positive SHAP values suggests that the associated feature tends to increase MCP, while blue points with negative SHAP values indicate a tendency to decrease MCP. A concentration of points around the zero axis implies that the feature has a negligible impact on the MCP. By analyzing these visualizations, we can gain valuable insights into the complex relationships between input features and the output (MCP). In this section, we leverage SHAP analysis to explain the impact of individual input features on the MCP. We prioritize the most influential features first, followed by a comprehensive examination of the effects of energy sources on electricity prices. Finally, we examine the contribution of market-related and financial features to the MCP. This systematic approach enables a thorough understanding of the complex interactions between various factors and their independent and collective impact on prices in the electricity market.

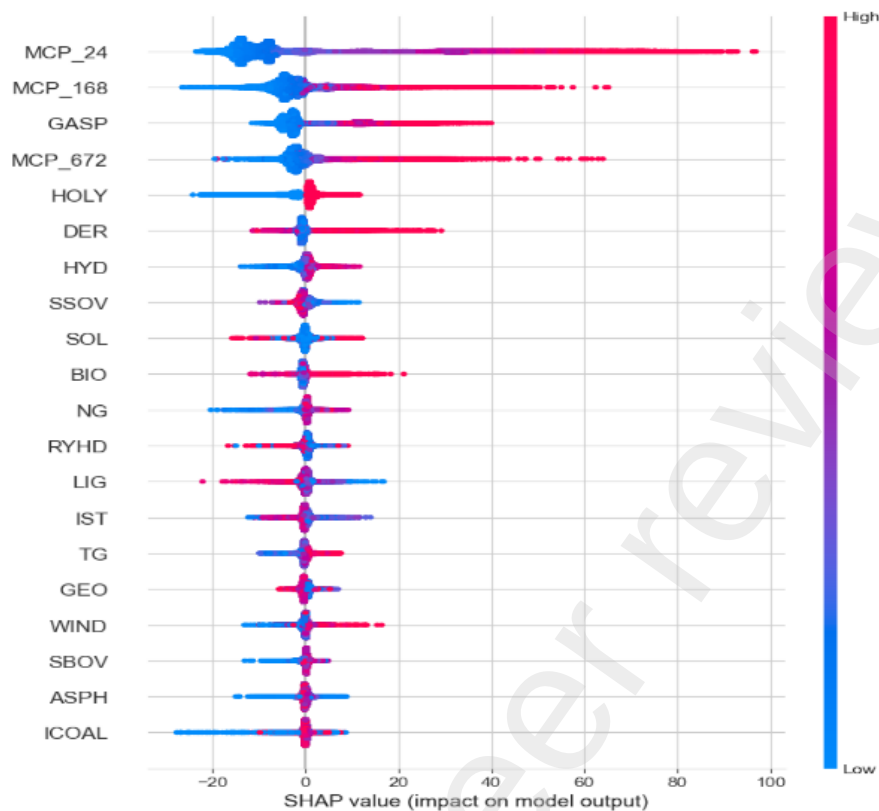


Figure 8. LightGBM Shap Value

SHAP values for the LightGBM model, illustrated in Figure 8, and those of other models (presented in Appendix E) consistently identify the MCP-24 variable, representing yesterday's price, as one of the most influential predictors of the day-ahead MCP. The SHAP analysis reveals a clear relationship, that is, low values of MCP-24 exert a strong negative influence on the MCP, while high values exert a strong positive influence. Among all features considered, MCP-24 stands out as the variable whose high values have the most substantial positive impact on the MCP. This implies that a high MCP-24 value significantly increases the likelihood of a higher MCP on the subsequent day. Conversely, a low MCP-24 value suggests a higher probability of a lower MCP the following day. Intermediate values, indicated by dark blue, purple, or dark red dots clustered around zero SHAP values, have a negligible effect on the day-ahead MCP.

Similar observations can be extended to MCP-168 and MCP-672, representing the MCP of the last week and the last month, respectively. Although their influence is slightly less pronounced than that of MCP-24 in the LightGBM model, the overall trend remains consistent. Low values of MCP-168 and MCP-672, indicated by light blue dots in the SHAP plots, exert a negative influence on the predicted MCP, suggesting a higher probability of lower prices on the following day. Conversely, high values of these variables indicate a higher likelihood of higher prices. Intermediate values, characterized by SHAP values near zero, have a negligible impact on the day-ahead MCP. Notably, MCP-24 and MCP-168, when exhibiting low values, emerge as two of the most influential features in decreasing the MCP, alongside day type (HOLY) and coal production (ICOAL).

These results are supported by the autocorrelation (ACF) and partial autocorrelation (PACF) analyses. ACF and PACF analyze the temporal behavior of average electricity prices in the market (Figure 9). These analyses assist in determining the extent to which past values of MCP influence current values.

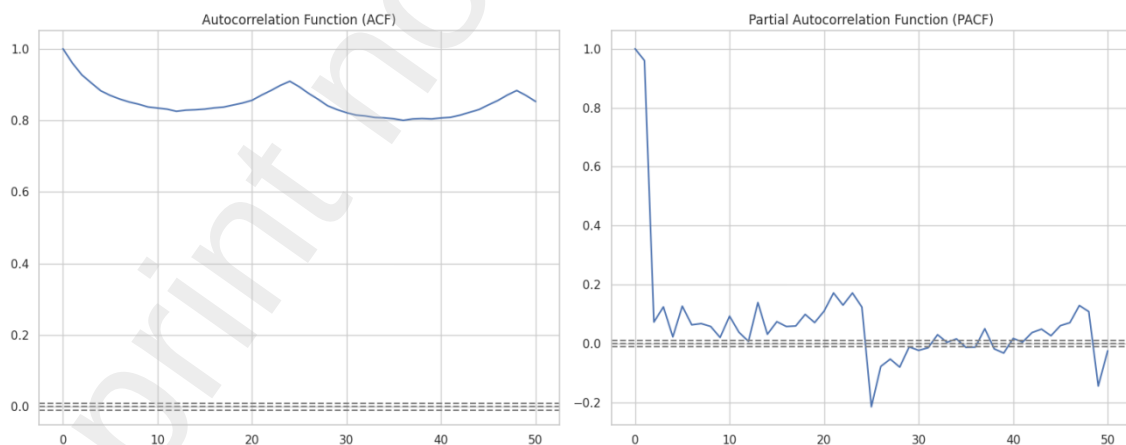


Figure 9. Autocorrelation (ACF) and Partial Autocorrelation (PACF) Analysis

The analysis of the ACF graph provides significant insights into the internal relationships within MCP data. The initial high values observed in the graph indicate a strong positive correlation with short-term past data of MCP, suggesting that past values can be crucial

benchmarks in current price predictions. Particularly, short-term historical data such as values from a few hours or a day ago appears to play a significant role in determining current MCP values.

The PACF graph analysis displays partial correlations between MCP and its past values. The prominent high values in the initial part of the graph indicate a substantial influence of short-term past values on current MCP values. However, the rapid decrease in these high values suggests a diminishing impact of past values over time. Specifically, the high PACF values for the first few lags emphasize the critical importance of early lags in predicting current MCP values, indicating a stronger relationship with recent past values. This observation suggests that historical data, especially from the first few hours or days, should be given more consideration.

The binary nature of the HOLY variable, representing weekdays (HOLY=1) and weekends/holidays (HOLY=0), presents challenges for independent evaluation due to its limited value range and inconsistent effects on the model. While a dense clustering of red dots near zero SHAP values suggests a minor upward influence of weekdays on the MCP, the wider distribution of blue dots associated with weekends/holidays along the x-axis indicates a more substantial but highly variable downward effect. This implies that not all weekends/holidays uniformly lead to lower prices. Consequently, the impact of the HOLY variable on MCP cannot be definitively ascertained in isolation. Its influence is likely contingent upon interactions with other variables within the model, necessitating a comprehensive analysis that considers the interplay of multiple factors to accurately assess its contribution to price fluctuations.

Regarding the daily exchange rate (DER), a dense distribution of red dots on the positive side of the SHAP plot reveals a strong positive correlation between high DER values and an increase in the MCP. This signifies that a higher exchange rate effectively drives up electricity prices on the following day. This relationship can be attributed to the substantial reliance of

Türkiye's electricity generation on imported fuels, such as natural gas (22%) and coal (17%), which collectively constitute nearly 40% of the total generation. As the DER increases, the cost of importing these fuels rises, leading to higher production costs and subsequently, higher electricity prices. Conversely, the scarcity of blue dots representing low DER values, along with their clustering around zero SHAP values, suggests a negligible impact of low exchange rates on the MCP. This indicates that even when the exchange rate is favorable, it does not significantly contribute to lowering electricity prices. This observation underscores the asymmetric impact of DER on MCP, where high exchange rates exert considerable upward pressure on prices, while low exchange rates have a limited downward effect.

An examination of energy generation sources reveals that the amount of electricity produced from both renewable and fossil fuels significantly impacts the MCP. To better comprehend this influence, it is essential to consider their respective shares in the overall electricity generation mix. Figure 10 illustrates the electricity generation shares as of 2022, providing valuable context for interpreting the impact of each source on the MCP. This analysis highlights the importance of considering the composition of the energy mix when analyzing price fluctuations in the electricity market.

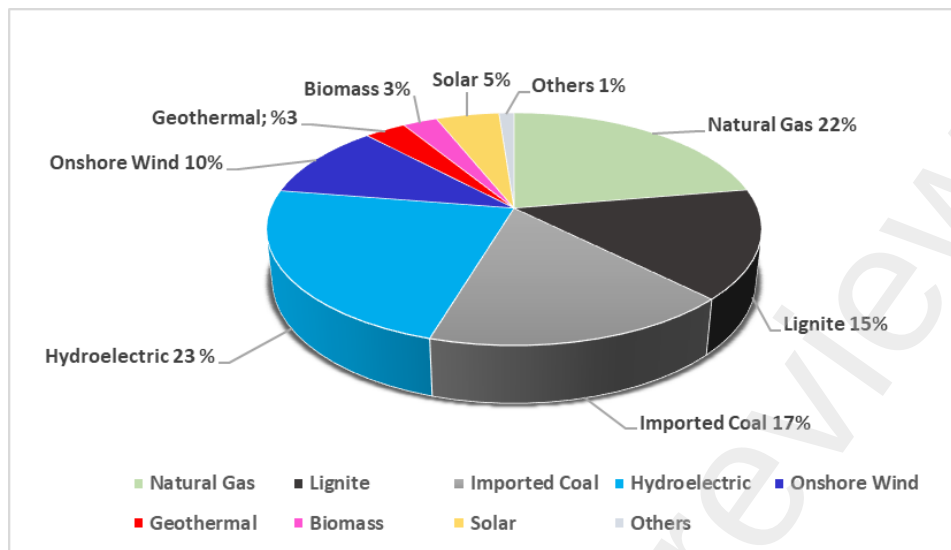


Figure 10. Turkish Electricity Generation Shares as of 2022

Contrary to initial expectations, the SHAP analysis reveals a positive correlation between high production from Dam Hydro (HYD) and the MCP, even though this effect is relatively minor in the LightGBM model. This observation is consistently replicated across all models (see Appendix E), with similar effect magnitudes observed in other well-performing models such as CatBoost (Figure E. 1) and XGBoost (Figure E. 2), lending further credence to this unexpected finding. This counterintuitive outcome may be attributed to the specific market dynamics and operational strategies of hydroelectric power plants in the Turkish electricity market. High hydroelectric production may coincide with periods of peak demand when the MCP is naturally elevated due to increased competition for limited supply. Alternatively, hydroelectric plants may strategically adjust their output to maximize revenue during periods of high prices, contributing to the observed positive correlation. Further investigation into the underlying mechanisms driving this relationship is warranted to gain a comprehensive understanding of the interplay between hydroelectric generation and price dynamics in the market.

Analysis of solar (SOL) generation reveals a nuanced impact on the Market Clearing Price (MCP). Low solar production, represented by blue dots clustered near zero SHAP values,

demonstrates a negligible effect on MCP. However, as solar generation increases, a more pronounced negative impact on MCP emerges, with a concentration of red dots on the negative side of the SHAP plot. This suggests that high solar production generally exerts downward pressure on day-ahead MCP, aligning with the expectation that increased renewable energy supply displaces more expensive generation sources, thereby lowering prices. This observation is corroborated by multiple models, including CatBoost, XGBoost, SVR (*Figure E. 4*), RF (*Figure E. 5*), and DT (*Figure E. 6*), which consistently demonstrate the negative correlation between high solar generation and MCP. Notably, the Sequential model deviates from this pattern, not considering solar generation as a significant predictor. This discrepancy may be attributed to differences in model architecture. Overall, the collective evidence suggests that solar generation, particularly at higher levels, plays a meaningful role in reducing the MCP. This finding underscores the potential of solar energy to contribute to lower electricity prices and promote a more sustainable energy mix.

In the case of biomass (BIO) generation, the concentration of blue dots at zero SHAP value signifies that low levels of biomass production have a negligible impact on the subsequent day's MCP. However, red dots are observed on both the negative and positive sides of the SHAP plot, with higher SHAP values on the positive side. This suggests that high biomass generation may slightly increase the MCP. This observation can be attributed to the relatively higher marginal cost of biomass energy compared to other renewable sources, as higher production costs can translate into increased prices in the electricity market. This pattern is consistent across most models, except for the RF model, which does not consider biomass generation as a significant predictor.

Natural gas price (GASP) emerges as a critical determinant of the MCP, exhibiting a strong positive correlation across all models. High gas prices consistently predict a significant increase

in the day-ahead MCP, reflecting the substantial influence of natural gas as a primary fuel source for electricity generation. However, the impact of low gas prices is less pronounced, generally demonstrating a negligible downward effect on the MCP. This asymmetry in the impact of gas prices is mirrored in the relationship between natural gas (NG) generation and the MCP. Low NG generation exerts a stronger downward pressure on the MCP compared to the slight upward pressure exerted by high NG generation. This observation, consistent across LightGBM, CatBoost, and XGBoost models, suggests that reduced reliance on natural gas for electricity generation can contribute to lower prices, particularly in scenarios where alternative sources are available. However, it is important to note that this effect is not uniform across all models, indicating potential model-specific nuances. Furthermore, the analysis reveals a complex interplay between natural gas prices and generation levels. While high gas prices reliably predict a high MCP, low gas prices combined with low NG generation do not necessarily guarantee a lower MCP.

While high generation from river hydro (RHYD) consistently demonstrates a lowering effect on the MCP across all models, the impact of low RHYD production remains insignificant. However, the RHYD variable itself does not consistently emerge as a significant predictor across all models, and the SHAP values exhibit considerable variability for both high and low generation levels. Consequently, the conclusions regarding the relationship between RHYD and MCP are not robust and should be interpreted with caution.

Although lignite coal is an important energy source for Türkiye with a 15% share in total electricity generation, only LightGBM, CatBoost, and RF models consider generation from lignite (LIG) a significant variable. High generation from lignite coal (LIG) strongly lowers the MCP and low generation increases the MCP according to LightGBM and CatBoost, the best-performing models. Similar interpretations can be drawn for generation from geothermal (GEO)

but with a lesser impact on the MCP. Only the SVR model considers it more significant and with a contrary impact, that is, high generation from geothermal strongly increases the MCP and low generation lowers the MCP.

Another interesting and unexpected result according to the LightGBM model is that high generation from wind (WIND) increases the MCP and low generation decreases it although the impact is not very substantial. CatBoost, SVR, and XGBoost models do not even consider the effect of the WIND variable. The sequential model does not give an implicit explanation of the effect. Only the DT model deduces that high generation from wind (WIND) decreases the MCP and low generation increases it.

High generation from import coal (ICOAL) shows no effect on the MCP, however, low generation has a strong decreasing effect, with a SHAP value of nearly 30 on the negative side, the most impactful among all parameters according to the LightGBM and RF models. The XGBoost model gives similar results for low generation but, according to this model, high generation from import coal (ICOAL) decreases the MCP. The CatBoost and SVR models do not consider ICOAL as a meaningful variable and the Sequential and DT models do not implicitly explain its effects.

The analysis of the impact of generation from black coal (BCOAL) on the MCP presents mixed results across different models. The LightGBM model does not consider the BCOAL variable as a significant factor in predicting MCP. In contrast, all other models include BCOAL in the MCP forecasts, indicating its potential relevance. However, the direction and magnitude of its impact vary. The CatBoost model offers the most coherent explanation. It suggests that high levels of BCOAL generation substantially decrease the MCP, aligning with the expectation that increased supply from lower-cost sources would drive prices down. Interestingly, the

reverse scenario, that is, low generation from black coal, does not appear to have a significant impact on the MCP.

Generation from fuel oil (FUEL) has a very low share in Turkish electricity generation (<1%), hence, the FUEL variable is not a significant factor for most of the models. LightGBM, CatBoost, and Sequential models do not include the variable in the forecast and the impact of the variable in other models is not considerable.

In addition to energy sources and historical MCP and gas prices, various market-related and climate variables influence the MCP forecast. These include trade value (TV), submitted sales order volume (SSOV), submitted bid order volume (SBOV), matched offers quantity (MO), price independent sales order (PISO), price independent bid order (PIBO), and hourly average temperatures of Ankara and Istanbul. However, the LightGBM model selectively incorporates only SSOV, SBOV, and the hourly average temperature of Istanbul (IST), while disregarding the remaining variables. The MO variable is only considered by the CatBoost model with a low impact.

Submitted Sales Order Volume (SSOV), representing the quantity of electricity supply bids offered to the market at the maximum clearing price step, is expected to exert downward pressure on the MCP as increased supply generally leads to lower prices. This relationship is corroborated by the SHAP analysis of LightGBM, CatBoost, SVR, and Sequential models, which consistently demonstrate that low SSOV values are associated with higher MCP. However, the converse effect, where high SSOV significantly lowers the MCP, is not as strongly supported by these models. Conversely, price independent sales order (PISO), representing the quantity of electricity supply bids offered to the market without any price step, provides complementary insights into the relationship between supply and the MCP. Despite not being included in the LightGBM model, SHAP analysis across all other models consistently

reveals a negative correlation between PISO and MCP, indicating that higher PISO values are associated with lower MCP. Consequently, a comprehensive analysis of the supply-side influence on MCP necessitates a joint consideration of both SSOV and PISO variables.

A parallel analysis can be conducted for price independent bid order (PIBO) and submitted bid order volume (SBOV), representing the quantity of electricity demand bids offered to the market without any price step and at 0 \$/MWh price step, respectively. However, surprisingly, the impact of these demand-side variables on the Market Clearing Price (MCP) is not as pronounced as that of their supply-side counterparts, SSOV and PISO. SHAP values of SBOV in LightGBM, SVR, Sequential, and DT models consistently indicate that lower SBOV values are associated with decreased MCPs, suggesting that reduced demand may lead to lower prices. Conversely, the effect of PIBO is less clear-cut, as it is represented only in the SVR model, where higher PIBO values are linked to increased MCPs. This disparity in the observed impact of demand-side variables compared to supply-side variables warrants further investigation. It is possible that other factors, such as price elasticity of demand, may be moderating the relationship between demand and MCP.

Trade value (TV), representing the hourly total financial volume of matching bids in the market, emerges as a significant predictor of the MCP in several models, despite not being considered by LightGBM. Specifically, CatBoost, XGBoost, and Sequential models indicate a strong positive correlation between high TV values and MCP, suggesting that increased trading activity and financial volume drive prices upward. However, the downward effect of low TV values on MCP is less pronounced in these models. While TV is also factored into SVR, RF, and DT models, its impact is less substantial and less clearly defined. The omission of TV from the LightGBM model may be attributed to the high impact of the MCP-24 variable, which represents the MCP of the previous hour. Given the inherent relationship between hourly total

financial volume and MCP, it is plausible that the MCP-24 variable indirectly captures the influence of TV within the LightGBM framework.

The hourly average temperature of Istanbul (IST) emerges as a relevant factor influencing the MCP in CatBoost, Sequential, SVR, and RF models. However, the precise nature of the relationship remains ambiguous. Notably, the CatBoost model reveals a discernible pattern wherein lower temperatures in Istanbul are associated with a slight increase in MCP, while higher temperatures correspond to a decrease. This observation can be attributed to the seasonal variation in electricity demand, particularly for heating purposes. During colder months, the increased need for heating, often fueled by natural gas, drives up electricity demand, thereby exerting upward pressure on the MCP. Conversely, warmer temperatures reduce the need for heating, leading to lower electricity demand and subsequently lower MCPs. In contrast, the hourly average temperature of Ankara (ANK) demonstrates a less pronounced and less consistent impact on MCP, as evidenced by its inclusion in only the SVR and DT models with relatively low significance.

6. Discussion and Conclusion

With the increasing liberalization and competition in electricity markets, the importance of explainable electricity price forecasting (EPF) has grown significantly for various stakeholders, including producers, consumers, trading companies, and investors. This study addresses this need by developing and analyzing models for forecasting Turkish day-ahead market prices. Employing a diverse range of machine learning models such as RF, DT, SVM, XGBoost, CatBoost, LightGBM, and Sequential ANN, this research utilizes Python programming for model construction and adheres to open-access principles by sharing the data and codes. Real-world hourly data from 2015 to 2023, sourced from the EXIST, the General Directorate of Meteorology, and the General Directorate of BOTAŞ, is meticulously preprocessed.

Hyperparameter optimization is performed for each model, resulting in substantial accuracy improvements. Model performance is evaluated using multiple criteria such as R^2 , MSE, MAE, RMSE, and MAPE. The LightGBM model consistently outperforms other models across these metrics, demonstrating its superior predictive capability. Furthermore, SHAP analysis is employed to provide valuable insights into the factors influencing electricity prices. This analysis enhances the interpretability of the models and aids in understanding the complex dynamics of the electricity market. The proposed prediction models, along with the accompanying SHAP analysis, offer a valuable tool for stakeholders in the electricity market. By simply updating the dataset, these models can be easily adapted to provide accurate and interpretable price forecasts, empowering decision-makers with actionable information. The following discusses the highlights of the study.

- Open access is vital for advancing EPF research. However, the lack of publicly available datasets and codes hinders reproducibility and progress in this field. The unavailability of data and codes impedes the replication of studies, contradicting scientific principles and hindering the evaluation and comparison of methodologies, thereby slowing down progress and innovation in EPF research. This study addresses this issue by sharing both the collected and preprocessed data, as well as the developed codes, to promote transparency and facilitate further research in EPF. By embracing open access, this study aims to contribute to the advancement of the field and foster a more collaborative and transparent research environment.
- Initial modeling attempts using the full dataset (2015-2022) yielded unsatisfactory results due to identified errors in many variables. These erroneous data points hampered model learning and accuracy. To rectify this, the dataset was refined to include only clean data from 2018-2022, resulting in a significant improvement in model accuracy. This finding highlights the critical importance of meticulous data preprocessing in EPF, as flawed or incomplete data can severely compromise model performance.
- Hyperparameter optimization, alongside data selection and preprocessing, plays a crucial role in enhancing the accuracy of electricity price prediction. Hyperparameters

govern the learning process and performance of machine learning models. By systematically experimenting with various model structures and parameter combinations, the optimal configuration can be identified, leading to improved prediction accuracy, generalization ability, and prevention of overfitting.

- Post-modeling analysis using SHAP values is crucial for understanding feature importance and their impact on electricity price predictions. SHAP analysis reveals the most influential factors, enabling a comparative evaluation of feature contributions and providing insights into the model's decision-making process. This enhances model transparency, interpretability, and reliability, ultimately leading to more informed decision-making in the electricity market.
- The rapid advancement of lightweight gradient boosting models, such as XGBoost, LightGBM, and CatBoost, has revolutionized forecasting research. These models offer high performance with reduced computational demands, providing significant time and cost savings for researchers working with large datasets. However, the choice of model and hyperparameter optimization process necessitates careful consideration of the trade-off between speed and performance, especially when computational resources are limited. This study highlights the practical challenges encountered during model selection and hyperparameter tuning, emphasizing the impact of these processes on research costs, timelines, and sustainability. Researchers are advised to anticipate potential financial and time constraints associated with lengthy training durations and, whenever feasible, prioritize parallel computing techniques and more efficient algorithms.

This study focused exclusively on the Turkish Day-Ahead Market, offering a comprehensive analysis of machine learning models' performance in this specific context. Future research can extend this work by incorporating larger datasets and comparing prediction accuracy and runtime across various electricity markets. Additionally, expanding the scope to include diverse variables and exploring alternative machine learning methods for regression analysis could further enhance our understanding of price dynamics. The insights gained from this study can serve as a valuable reference for future investigations into EPF. By diversifying

input variables, classifying different periods for prediction, and employing hybrid methods, researchers can strive to achieve even better prediction accuracy. Moreover, the evolving landscape of electricity markets, characterized by the growing integration of wind and solar power, presents new research opportunities. Future EPF models will need to account for the increasing uncertainty associated with renewable energy production to maintain their predictive power in this dynamic environment.

Acknowledgment

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Appendix A

Abbreviations

ACF	Autocorrelation
AI	Artificial intelligence
ANFIS	Adaptive-Network-based Fuzzy Inference System
ANN	Artificial Neural Network
AR	Autoregressive Model
ARIMA	Autoregressive Integrated Moving Average
ARMA	Autoregressive Moving Average
ARX	AutoRegressive with eXogenous
BI	Best Individual
BPM	Balancing Power Market
CMARS	Multivariate Adaptive Regression Splines
CNN	Convolution Neural Network
DAM	Day-ahead Market
DTR	Decision Trees Regression
DL	Deep Learning
EEX	European Energy Exchange
EMOIC	Energy Markets Operation Inc.
EPF	Electricity Price Forecasting
EXIST	Exchange Istanbul
FNN	Feed-forward Neural Network
GDM	General Directorate of Meteorology
GRU	Gated Recurrent Units
IDM	Intra-Day Market
LightGBM	Light Gradient Boosting Machine
LSTM	Long Short-Term Memory
LSTNet	Long- and Short-Term Time-series Network
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MCP	Market Clearing Price
ML	Machine Learning
MSE	Mean Squared Error
MLR	Multiple Linear Regression
Nord Pool	Nordic Power Exchange
PACF	Partial Autocorrelation
PJM	Pennsylvania-New Jersey-Maryland Interconnection
PPC	Petroleum Pipeline Corporation
RF	Random Forest
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
SARIMA	Seasonally Autoregressive Integrated Moving Average
SETAR	Self-exciting threshold auto-regressive regression
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machines
XGBoost	Extreme Gradient Boosting

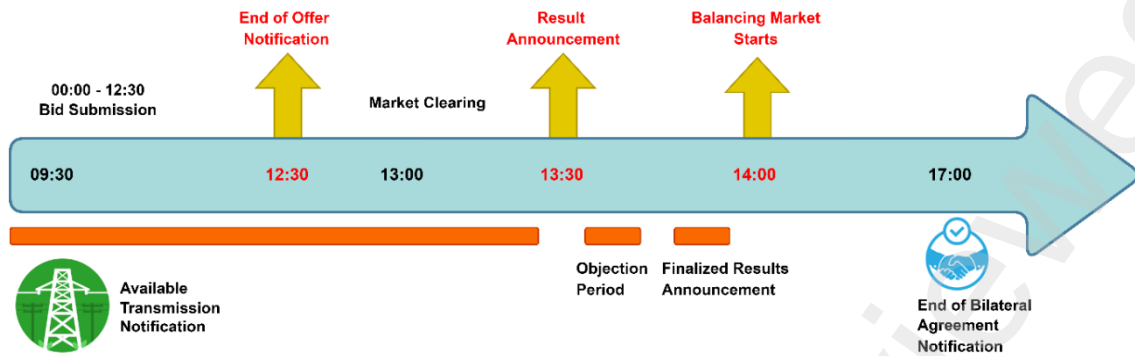


Figure A.1. Turkish Day-Ahead Market Process

Table A. 1. MCP Forecast Studies in the Turkish Day-Ahead Market

Authors (Date)	Methodology	Variables Used	Period Covered	Results (Error Metric Values)
Şenocak and Kahveci [75]	ANN, ANFIS	The production quantities of natural gas, hydro, and natural gas-liquid fuels-lignite energy resources	2010- 2015	MAPE, ANFIS
Demirezen and Çetin [76]	Random Forest, Regression Support, Vector Regression	One day before MCP, a week ago MCP, a month ago MCP, day of the week (Weekdays 0; Weekends 1), holiday (holidays and special days 1; other days 0), MCP one hour ago, MCP two hours ago, MCP three hours ago, trade volume	01.01.2019-10.03.2020	MAE, MAPE, RMSE
Ertaylan et al. [77]	Artificial Neural Network	Date, Time, One day before MCP, A week ago MCP, Total consumption amount, Total production amount, Gas reference price Holidays and Ramadan days (Independent variable)	01.01.2019-31.12.2019	MSE
Akay et al. [78]	LSTM, SVM	System Marginal Price	01.01.2018-30.06.2021	MAPE
Arslan and Ertuğrul [79]	Multiple Regression, Artificial Neural Networks	Energy production quantity, Demand quantity	2020 (March, April, May, June)	MAPE
Karatekin and Başaran [80]	Linear Regression, Polynomial Regression, Artificial Neural Networks, Xgboos	Day-ahead hourly Market Clearing Price (MCP), Hourly production for each electricity generation source	2015-2020	RMSE
Sirin and Yılmaz [81]	Quantile regression	Demand, MCP, Wind, River Hydro, Gas	May 2016 – May 2019	least squares error rate
Avcı et al. [63]	Time series, Artificial neural network, Seasonality models	Forecasted demand, Margin, Day-specific dummy, Holiday dummy	01.01.2012 – 31.12.2015	MAPE

Tayşi et al. [64]	SARIMA, ANN	One day ago MCP at the same hour, One week ago MCP at the same hour, Average MCP in the last 24 hours	December 2012 – November 2014	MAPE
Uğurlu et al. [62]	Markov, Navie, SETAR, ANN, CNN, SARIMA, LSTM, GRU	24-hour delayed price, 168-hour delayed price, 1-hour delayed price, 48-hour delayed price, 23-hour delayed price, 72-hour delayed price, 336-hour delayed price, Demand forecast instead of supply, Temperature, Demand/supply realized with a 24-hour delay, Balancing of the market price with a 24-hour delay	01.01.2013 - 21.12.2016	MAE
Özyıldırım and Beyazıt [59]	Linear Regression, Radial basis function	MCP lags, Temperature, Square of temperature, Hourly, Daily, Monthly, Holiday dummies	2010 - 2013	MAE, MSE
Kölmek and Navruz [61]	ANN, AR	Historical day-ahead prices, Demand forecast Average temperature, Consumption forecast, Amount of bilateral contracts, Day code	December 2009 – November 2010	MAPE
Avni Özözen et al. [65]	SARIMA, ANN	Day-ahead hourly Market Clearing Price (MCP), Demand/ Supply	2014 - 2015	MAPE, MSE
Arifoğlu and Kandemir [82]	MLP, CNN, LSTM	Days of the week and public holidays, Real-time consumption, Hydroelectric power generation, System marginal price, Load forecast plan, Wind speed, Temperature	13.06.2016 – 28.03.2021	MAPE

Appendix B

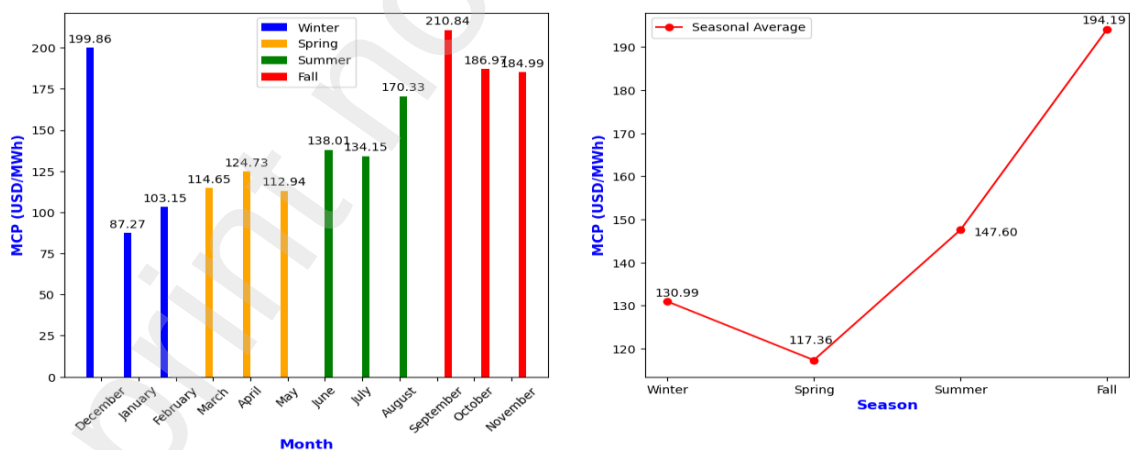


Figure B. 1. Average monthly and seasonal electricity prices in Turkish Day-Ahead Market in 2022

In the electricity market, prices can vary above or below the annual average during specific hours. As seen in Figure B. 2, the average MCP is above the annual average between 00:00-08:00, 11:00-14:00, and 14:00-22:00. However, between 01:00-07:00, 12:00-13:00, and at

23:00, the MCP is below the annual average. As depicted in Figure B. 3, on weekdays, prices are relatively consistent, while weekend prices exhibit variability. Saturday morning prices are similar to other weekdays, with a decrease observed on Saturday night and Sunday.

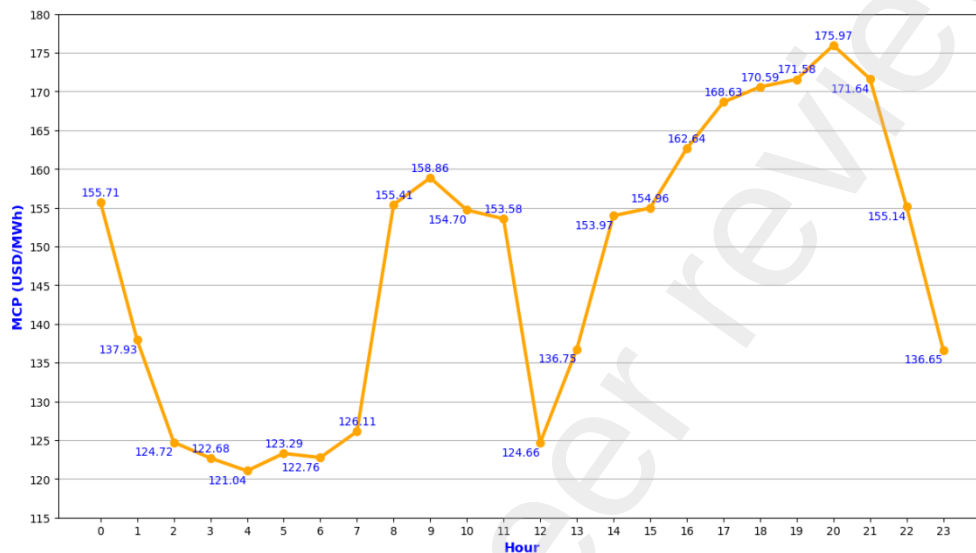


Figure B. 2. Average hourly electricity prices in the Turkish Day-Ahead Market in 2022

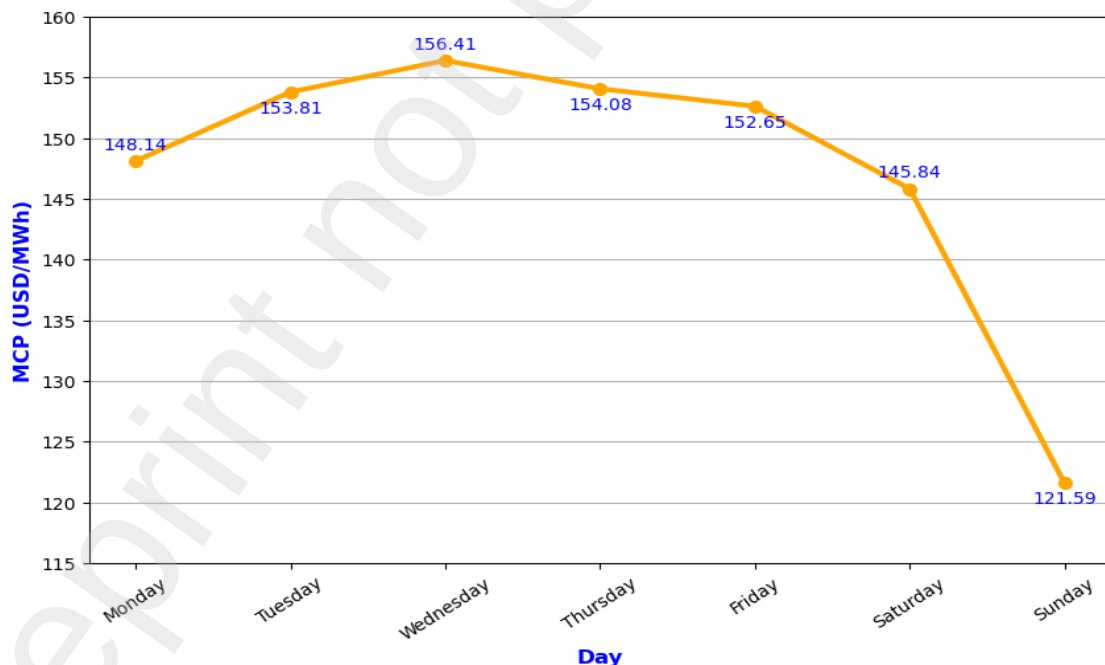


Figure B. 3. Average Daily Turkish Day-Ahead Market electricity prices in 2022

Figure B. 4 shows time series lines depicting daily and weekly average MCP values. Daily data exhibits more fluctuations, while weekly data smoothens these fluctuations, reflecting

more general trends. There is a general upward trend from 2018 to 2023, indicating an increase in prices in the day-ahead electricity market during this period.

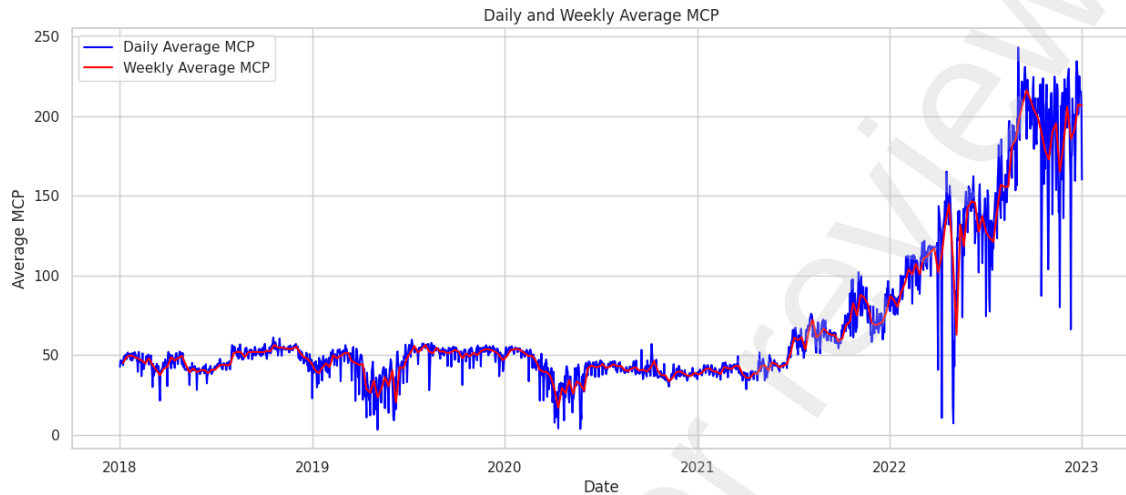


Figure B. 4. Average Daily and Weekly Turkish Day-Ahead Market electricity prices

Time series components are decomposed into three elements: trend, seasonality, and residuals (Figure B. 5). The trend component illustrates the long-term tendency in the data, showing a slowly increasing trend over time. The seasonality component displays regular fluctuations throughout the year, even though there might not be a distinct annual pattern. The residuals component represents variations that cannot be explained by trend and seasonality. These variations are typically random and irregular.

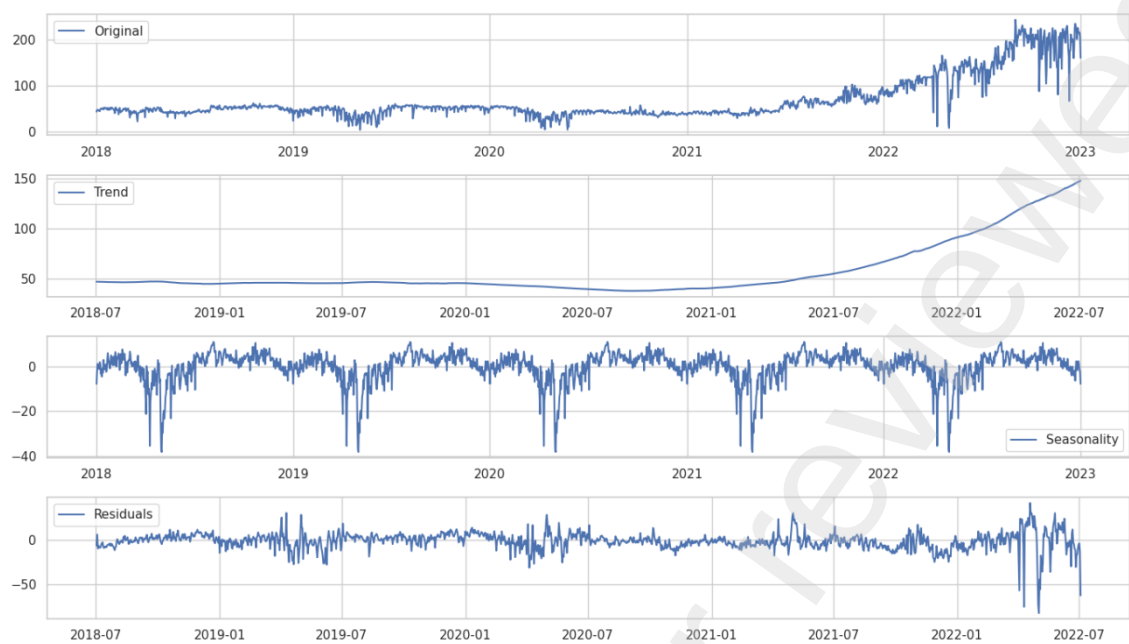


Figure B. 5. Original, Trend, seasonality, Residual

Appendix C

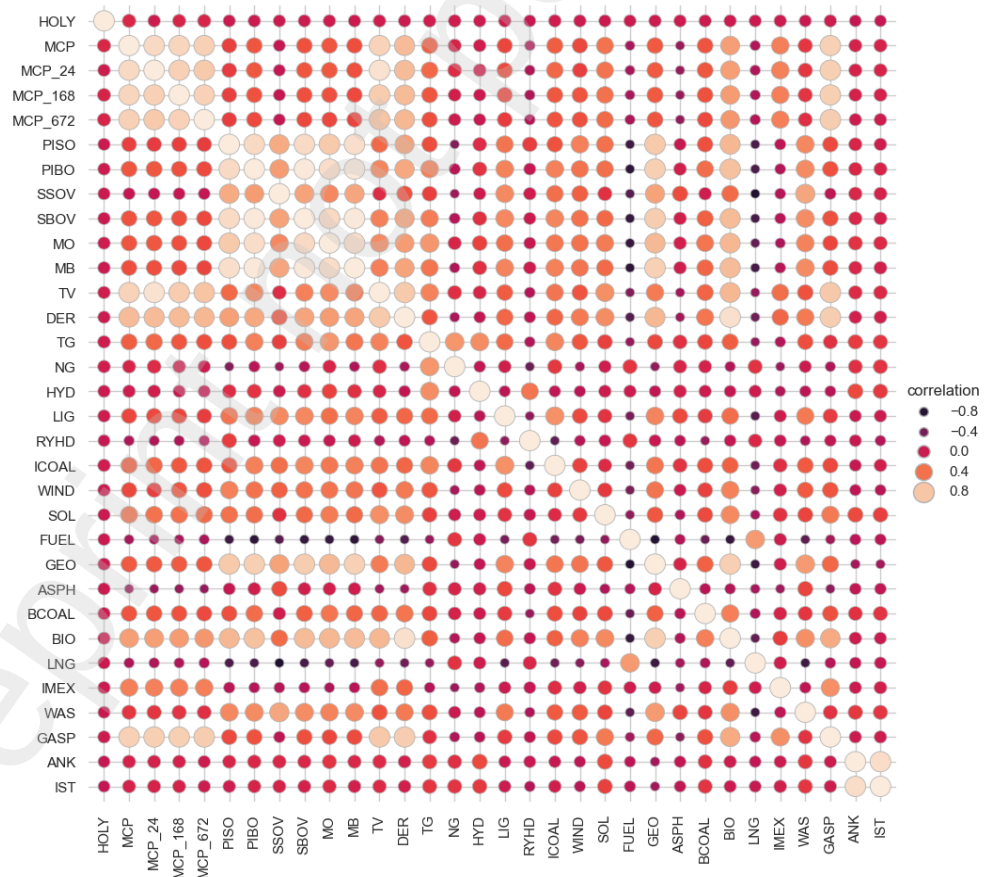


Figure C. 1. Correlation Heat Map

Table C. 1. Model hyper-parameters

XGBoost	learning_rate: Controls the model's adaptation speed to new information	0.05
	max_depth: Determines the depth of the decision tree	11
	n_estimators: Sets the number of decision trees	150
	colsample_bytree: Specifies the ratio of randomly selected features for each decision tree	0.5
	objective: Determines the error criterion that the model is trying to optimize	reg:squarederror
	alpha: Controls the amount of L1 regularization (Lasso)	10
	Train MAE	%3
	Test MAE	%9
	Train RMSE	%4
	Test RMSE	%17
RF	n_estimators: Determines the number of decision trees	300
	max_depth: Specifies the depth of the decision tree	20
	max_features: Determines the number of randomly selected features for each decision tree	20
	Train MAE	%4
	Test MAE	%9
	Train RMSE	%7
	Test RMSE	%19
SVR	C: Regularization coefficient that affects the model's complexity and generalization ability	900
	Epsilon: Determines the error tolerance level	7
	Train MAE	%9
	Test MAE	%10
	Train RMSE	%15
	Test RMSE	%17
DT	max_depth: Specifies the depth of the decision tree	10
	criterion: Splitting criterion (e.g., Gini or entropy)	squared_error
	max_features: Determines the number of features considered at each split point	15
	Train MAE	%9
	Test MAE	%12
	Train RMSE	%16
	Test RMSE	%24
CatBoost	iterations: Number of training iterations	500
	depth: Specifies the depth of the decision tree	7
	learning_rate: Controls the speed of the model's adaptation to new information	0.3
	loss_function: Determines the error criterion that the model is trying to optimize	RMSE
	silent: Suppresses information messages during training	True
	Train MAE	%6
	Test MAE	%9
	Train RMSE	%9
	Test RMSE	%17
LightGBM	learning_rate: Controls the speed of the model's adaptation to new information	0.2
	num_leaves: Specifies the number of leaf nodes in the decision tree	60
	n_estimators: Determines the number of decision trees	350
	Train MAE	%3
	Test MAE	%8
	Train RMSE	%4
	Test RMSE	%16

Sequential	<pre> model = Sequential() model.add(Dense(64, activation = 'selu', input_dim = X_train.shape[1])) model.add(Dense(2048, activation = 'selu')) model.add(Dense(1024, activation = 'selu')) model.add(Dense(512, activation = 'relu')) model.add(Dense(256, activation = 'selu')) model.add(Dense(128, activation = 'relu')) model.add(Dense(64, activation = 'selu')) model.add(Dense(32, activation = 'relu')) model.add(Dense(16, activation = 'selu')) model.add(Dense(8, activation = 'relu')) model.add(Dense(1)) opt = Adam(learning_rate = 0.001) model.compile(optimizer = opt, loss = 'mse') </pre>	Dense: Represents hidden layers, specifying the number of neurons and activation function Adam: Optimization algorithm that controls the learning rate Mse: Error function that measures the model's prediction errors.
	Train MAE	%7
	Test MAE	%9
	Train RMSE	%12
	Test RMSE	%18

Appendix D

Machine learning models possess the capability to solve a variety of problems, yet the performance of each model can vary depending on the data, the type of problem, and the complexity of the model itself. Therefore, during the model selection and tuning process, it is essential to assess how well the model performs on the data. Various metrics and visualization techniques are employed to evaluate the model's performance. Among these are graphs that illustrate the relationship between the actual values and the values predicted by the model. Such graphs are used to visually express how well the model captures patterns in the dataset and the accuracy of its predictions. In this study, an analysis is conducted to compare and evaluate the prediction performance of seven different machine learning models. Each model is trained on the same dataset and made predictions. The analysis aims to visualize the prediction errors of the models and to determine which model best solves the problem by comparing them. For this purpose, graphs that reflect the relationship between the actual values and the values predicted by the models are presented in Figure D. 1 to Figure D. 6. Actual values are shown on the x-axis, while predicted values are on the y-axis. Green dots represent the prediction made by the model for each data point. The red line illustrates an ideal scenario where the model would

make perfect predictions. The black line represents the curve that best fits the training data according to the model. The closer the black and red lines are, the more accurate the model's predictions are.

The Sequential model (Figure D. 4) and SVM (Figure D. 3) demonstrated average performance, lagging behind the high-performing models but still delivering satisfactory results. However, the Random Forest (Figure D. 2) model exhibited lower performance than these two models. The lowest performance was shown by the Decision Tree model (Figure D. 5), which made the least accurate predictions in EPF. Ultimately, the LightGBM model stands out as the best-performing model compared to all others, making it the most suitable model for electricity price prediction.

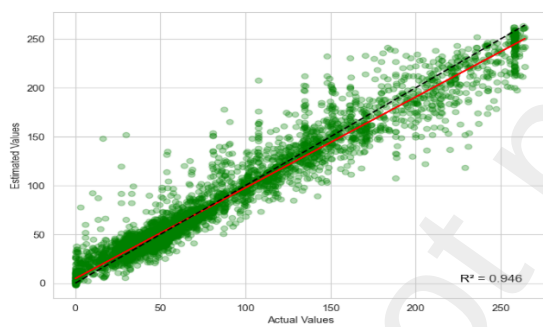


Figure D. 1. Prediction Error for XGBoost

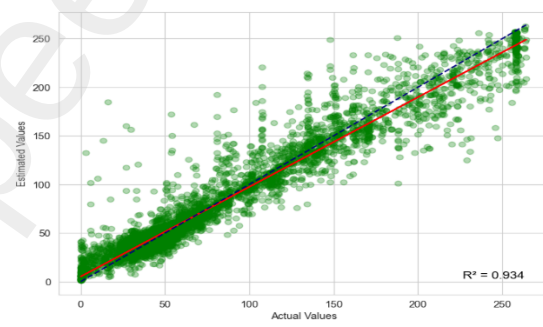


Figure D. 2. Prediction Error for RF

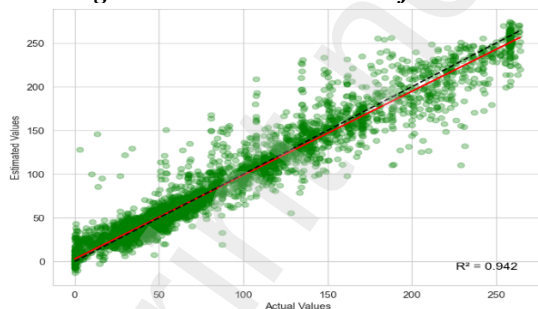


Figure D. 3. Prediction Error for SVR

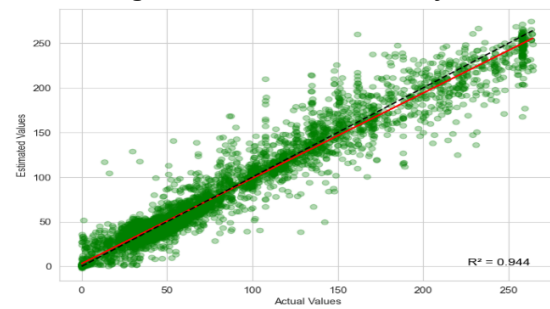


Figure D. 4. Prediction Error for Sequential

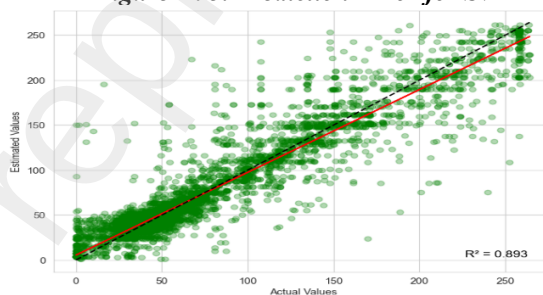


Figure D. 5. Prediction Error for DT

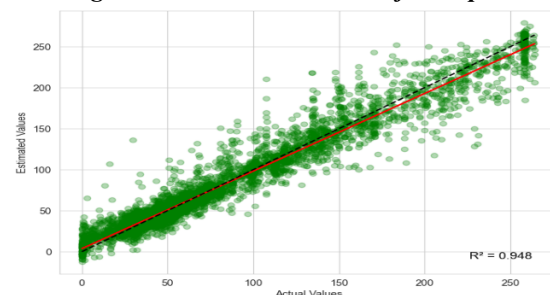


Figure D. 6. Prediction Error for CatBoost

Appendix E

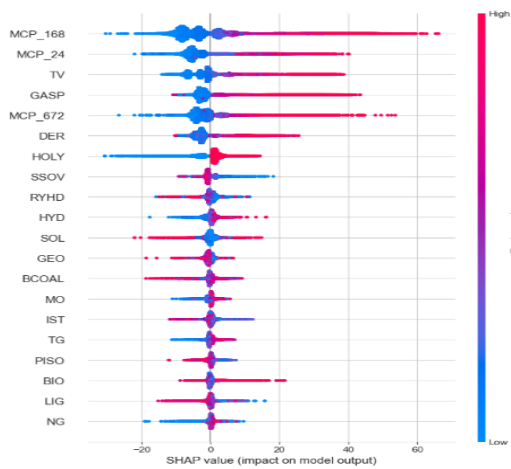


Figure E. 1. CatBoost Shap Value

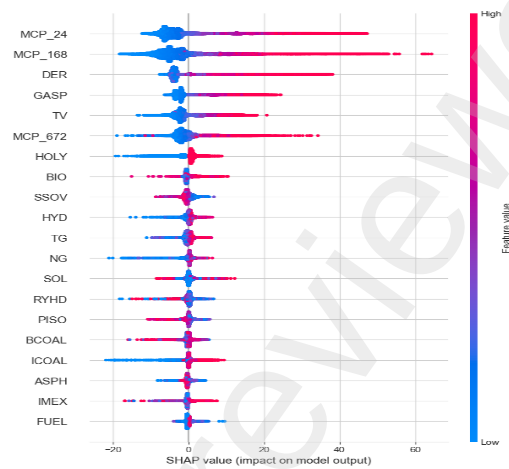


Figure E. 2. XGBoost Shap Value

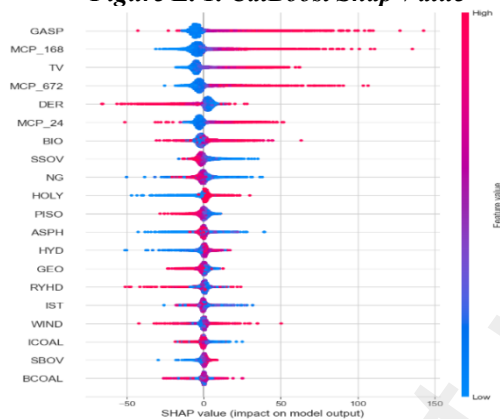


Figure E. 3. Sequential Shap Value

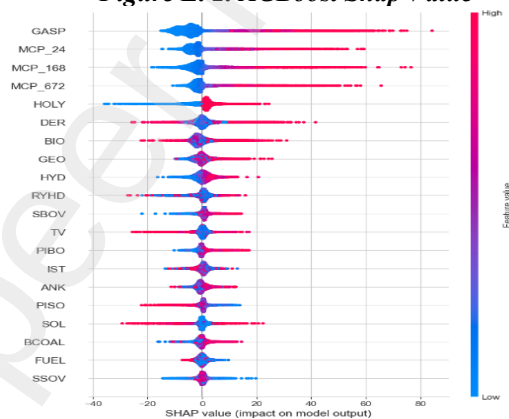


Figure E. 4. SVR Shap Value

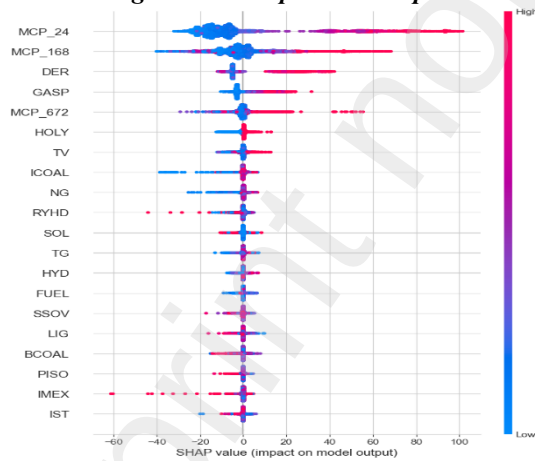


Figure E. 5. RF Shap Value

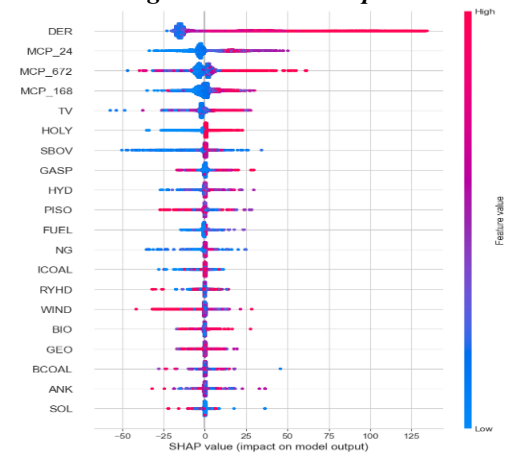


Figure E. 6. DT Shap Value

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